List of Corrections

I firstly demonstrate use cases where the spatial analysis of stereo audio has the opportunity to solve real-world problems in audio transmission, dissemination, and reproduction. Accordingly, I gather existing methods to describe stereo image and assess their abilities to detect the spatial aspects of a variety of musical audio sources. 12
Do we need to show the effects track? 25
Unfortunately I cannot demonstrate it in this thesis as Tardieu et al.'s method produced spurious results 41
These aren't stereo features. Maybe mention that. 47

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Abstract

Spatial information is crucial to the listener's experience of music, and stereophonic recordings have long been the standard for capturing this spatiality. Recent advancements in computing, coupled with the availability of new, very large musical corpora, have created opportunities for large-scale quantitative analysis of the stereo image of recorded music. In this licentiate thesis I address a key gap in current research: direct analysis of raw music recordings to describe the stereo image. I explore existing methods, assess their effectiveness, and demonstrate potential use cases for spatial analysis within audio reproduction, transmission and musical performance.

Dedicated to the one guy at Uppsala University who made a latex template 14 year ago

List of papers

This thesis is based on the following papers, which are referred to in the text by their roman numerals.

I A. Franklin et al., 'Development of a Web-Based Real-Time Distributed Reverberation Chamber for Live Concert Applications,' *Journal of The Audio Engineering Society*, vol. 73, no. 12, pp. 872–881, Mar. 2025. DOI: 10.17743/jaes.2022.0245

II R. Mitic and A. Rossholm, 'A Systematic Literature Review of Methods to Describe the Stereo Image of Recorded Music,' *Journal on Audio, Speech, and Music Processing*, vol. 2026, no. 1, p. 29, 9th Jun. 2026, ISSN: 3091-4523. DOI: 10.1186/s13636-026-00463-4

III R. Mitic and A. Rossholm, 'Towards an objective comparison of panning feature algorithms for unsupervised learning,' in *Proceedings of the 28th International Conference on Digital Audio Effects (DAFx25)*, Ancona, IT, 2nd Sep. 2025. [Online]. Available: https://dafx.de/paper-archive/2025/DAFx25_paper_26.pdf

IV R. Mitic et al., 'Perceptual Evaluation of Binaurally Panned Speech and Music Mixtures,' *Acta Acustica*, in review 2026

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1. Introduction

It is a widely accepted truth in the world of music production and recording that spatial information in audio is crucial to listeners' sense of immersion and connection with the music. To capture that spatiality, stereophonic recording and reproduction has been the standard format since it gained popularity in the 1950s [5]. The spatial analysis of subsequent decades worth of stereo recordings presents a unique opportunity to study human musicality, as each recording reflects and encodes the creative choices made by the musicians and recordists involved.

Within the music community it is common to speak about spatiality in a qualitative sense, using terms such as 'stereo image' and 'stereo field' [1] to encompass all audible spatial aspects. In academia however – and specifically in the Music Information Retrieval (MIR) community – there have been several explicit attempts to translate these discussions into a quantitive domain [6–8]. All were performed in an era when musical datasets were hand-curated/-labelled, fairly small (1,000 - 2,000 tracks), and, most notably, private. In recent years new musical corpora have become available, e.g. MTG-Jamendo [9] and Free Music Archive (FMA) [10], which are both orders of magnitude larger than their predecessors, containing approximately 55,000 and 100,000 tracks respectively. In addition, computing techniques in Machine Learning (ML) and Deep Learning (DL) have progressed to the point where it is now possible to learn extremely high-level features on these vast datasets, thereby making sense of the human creativity represented in the recordings.

A typical use case for spatial descriptors that has emerged both in academia and industry is classification, and in particular, auto-tagging of musical genre/performer [11, 12]. Such features can be used in music similarity and recommendation tasks [13, 14]. It has also been shown that stereo field classification of subsections within a larger work can act as a precursor to other tasks such as segmentation, audio enhancement, upmixing, and source separation [15–17]. The latter is very relevant to the audio distribution industry, which is currently experimenting with newer audio formats such as ambisonic and object-based audio. While these new formats offer exciting opportunities for both distributors and listeners, there remains a huge back-catalogue of stereophonic recordings that still need to be playable on modern reproduction equipment. A detailed understanding of their spatial content would therefore be beneficial.

[1] The terms 'stereo image' and 'stereo field' are freely interchangeable and, as is normal in the music production community, I use them both throughout this thesis.

Historically, binaural audio (which is a subclass of stereo) has been of particular interest because its relevance to human hearing. Thus, it is extremely widely studied, and the underlying fundamental techniques are well-established [18].

Yet there remains a gap in current research regarding direct analysis of raw music recordings, and in particular, describing the stereo image of said recordings at a high level. Through the work described in this thesis, I aim to fill that gap. FiXme: I firstly demonstrate use cases where the spatial analysis of stereo audio has the opportunity to solve real-world problems in audio transmission, dissemination, and reproduction. Accordingly, I gather existing methods to describe stereo image and assess their abilities to detect the spatial aspects of a variety of musical audio sources.

2. Research Questions and Contributions

The underlying purpose of the research documented in this thesis relates to MIR, and in particular to generating an understanding of the links between human musical creativity and musical output. While it might be possible for the creative people involved to explain how they make their music, it is far more objective to analyse their output as a representation of their musical ideas.

2.1 Research Questions

Given the aforementioned context, the overarching research question (RQ) can be formed as:

How do music producers use stereo field?

The basis of the research is access to extremely large musical corpora and modern computing techniques. The number and variety of recordings available for analysis makes them a decent representation of music producers in general. The spatial aspects of each track can be analysed individually using traditional Digital Signal Processing (DSP) techniques and a population-level analysis can be performed on the spatial features of all tracks together. Thus, I have broken down the main RQ into two smaller ones:

RQ_1: *How does the stereo image of a recording relate to real-world musical knowledge?*

RQ_2: *What methods can be used to describe the stereo image of a music recording?*

These concrete questions are addressed through the four papers included in this licentiate thesis. RQ_1 provides an important context of what quantitive spatial aspects receive attention from music practitioners, while RQ_2 establishes a state-of-the-art baseline of relationships between those quantitive descriptions and objective measurements. The overarching research question spans both licentiate and doctoral studies, with future plans being detailed in chapter 7.

2.2 Research contributions

This section presents a summary of the research contributions achieved by the 4 included papers and explains their relationship to the RQs. Many parts of the research required spatial descriptors to be extracted from raw audio signals. To facilitate this I implemented a library of software tools, allowing all the research tracks to be linked by a common mathematical framework.

2.2.1 An understanding of the musical uses of stereo field

With respect to the overarching research question, the act of personally creating musical output gives an insight into how music producers in general might use stereo field. We initially embarked on such a project, using audio from a remotely-located stereo microphone as the primary sonic material in a set of musical performances. Paper I was a retrospective account of the process, providing technical information to any musicians wishing to make a similar performance in the future. The paper also provided suggestions for future iterations of similar instruments and the standardized Application Programming Interfaces (APIs) which are built into web browsers, giving a basis for future work.

The software library presented the opportunity to apply a more technical approach to RQ_1. The exploration of stereo field in existing recordings furthered the knowledge gained from an individual musical project by allowing us to analytically compare stereo fields created by many producers. This allowed us to establish an understanding of the range and variation that stereo field can embody.

2.2.2 Signal processing techniques relating to stereo field

The software implementation and the resulting paper III contribute an understanding of how some established techniques for measuring stereo field react to existing recorded music. Moreover, that understanding is not limited to the techniques specified in paper III since I developed the software continuously throughout several literature searches, resulting a full-featured library that includes many other spatial analysis techniques.

In order to fully answer RQ_2, an extensive Systematic Literature Review (SLR) (paper II) was performed with two intentions. First and foremost, the intention was to conclusively find all existing mathematical techniques that describe stereo field. In addition, there was a goal to find 'cross-pollination' opportunities from disparate areas of audio research which nonetheless derive spatial cues from audio signals. In doing so, it was possible to form a 'general model' of Stereo Image Profiles (SIPs), thereby making it easier to for future researchers to find or create a SIP that suits their specific use case.

2.2.3 Psychoacoustics of mixed music and speech

The perceptual listening test described in paper IV provided a new perspective on a classic problem in psychoacoustics research. While there is extensive literature describing how people perceive music or speech that has been intentionally masked by another sound, our experiment used music and speech that were intended to complement each other. The results firstly suggested that while people prefer speech that is panned towards the centre, there was an opportunity to increase the audible pleasantness by using very subtle panning, and moreover that said panning might be automated by analysis of the stereo field (thereby helping to answer RQ_1). Furthermore, it became clear that the intelligibility of speech was paramount to pleasantness, despite the original assumption that the music and speech complemented each other.

2.2.4 Setup for future work on large datasets of audio

In regard to the future work described in chapter 7, the SLR provided both a list of relevant audio datasets, and a many standardized implementations of mathematical techniques that extract spatial information from audio. Paper III also presented a good opportunity to assess the computational complexity of the algorithms implemented in the software library, thereby highlighting some opportunities for optimization when they are applied to extremely large corpora.

3. Background

This chapter provides the fundamental background knowledge that underpins the work included in the thesis.

Music production techniques related to stereo field are thoroughly rooted in the human perception of sounds, exploiting well-understood auditory phenomena for creative means. However, that understanding is a fairly recent development. In fact, musical practice involving stereophonic sound predated the scientific knowledge of perception by several decades, with one of the earliest known examples dating back the late 19th century [19]. It wasn't until mid 20th century that the work of Fletcher and Munson [20] and Jeffress [21] outlined the psychoacoustic principles that music production would eventually exploit. In this chapter we draw a line from the basic human physiology of the auditory system, through psychoacoustics, DSP theory, and music production, eventually linking them all to the modern computing techniques used as the basis for the work detailed in this thesis.

3.1 The mechanics of hearing and auditory perception

The physical characteristics of the human auditory system govern what we as humans are able to perceive. In particular, our perception of frequency, loudness, and directionality are key when assessing stereo field.

3.1.1 Physiology of the inner ear

The ear contains three major sections, loosely labelled the 'outer', 'middle', and 'inner' ear (see fig. 3.1). When air vibrations (i.e. sound waves) flow into the outer ear, mechanical coupling in the middle ear transmits vibrations to the cochlea – the major organ of the inner ear. The cochlea is a conical tube which is coiled up into a spiral, filled with an incompressible fluid, and covered on either end by a membrane. Through the centre of the tube runs the basilar membrane, which is a covered in hairs named cilia. When vibrations are mechanically transmitted to the base of the cochlea, the internal fluid responds by transmitting those vibrations to the basilar membrane, and the cilia will bend and twist accordingly. The cilia are linked to nerve fibres, and will initiate electrical pulses when manipulated. These electrical pulses are sent to the brain and interpreted as sounds [22].

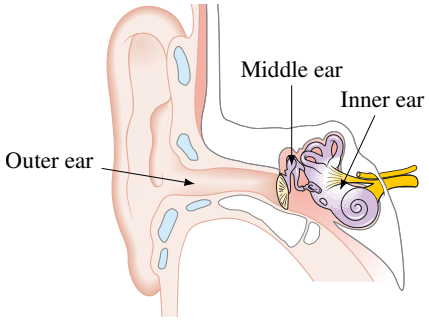


Figure 3.1. Diagram of the human ear*. Air flows through the outer ear and vibrates the ear drum. The middle ear mechanically couples the outer ear to the inner ear. The inner ear converts mechanical movements to electrical signals.

*Original image by Cancer Research UK is licensed under CC BY-SA 4.0. Annotations have been edited.

The basilar membrane reacts to a pressure wave from an incoming sound in an elastic way, with a buldge travelling from the base (the widest section) to the apex (the narrowest section). That buldge will change amplitude as it travels due to the basilar membrane's non-uniform width and stiffness, and the distance at which is reaches its maximum amplitude depends on the frequency of the incoming sound. Thus, the cilia at different positions along the cochlea have different 'critical frequencies' that they respond to. The critical frequency increases logarithmically as one moves from the base to the apex, as shown in fig. 3.2. The cochlea therefore acts to convert time-domain sound waves into time- and frequency-dependent electrical excitations, enabling us to perceive different frequencies simultaneously.

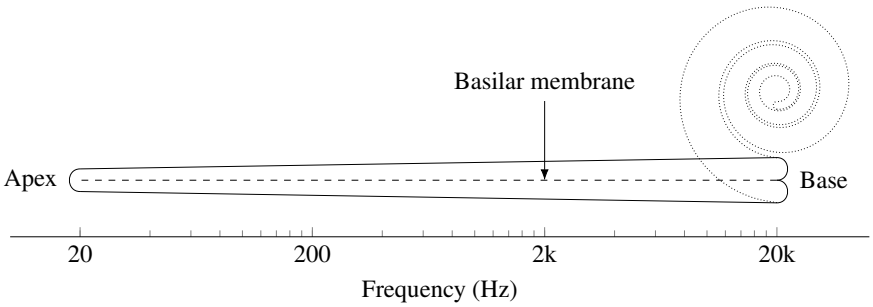


Figure 3.2. Tonotopic diagram of the human cochlea. Dotted lines represent the cochlea in its normal shape while solid lines show a hypothetical unfurled cochlea with the basilar membrane running down the centre shown as a dashed line. The frequency response of the basilar membrane changes logarithmically along its length, with high frequencies detected at the base and low frequencies at the apex.

3.1.2 Directional hearing

Section 3.1.1 explains how sound waves are translated in to electrical impulses inside a single ear. When this mechanism is expanded to both ears, combinations of the two sets of electrical impulses can be combined to determine the direction of an incoming sound.

The psychoacoustic study of spatial hearing has been an active area of research since the mid 20th century, with early examples including the Jeffress model in which time differences between sound events at each ear contribute to a person's ability to locate sounds in space [21]. The work of Lindemann a few decades later expanded on this idea by translating the earlier theory to a DSP context, and establishing the Interaural Level Difference (ILD), Interaural Time Difference (ITD), and Interaural Cross Correlation (IACC) as important measures to describe the spatial aspects of binaural signals in a quantifiable way [23].

Put simply, we are able to perceive the incoming direction of a sound by observing both the amplitude difference and time difference of coincident sound waves arriving at the left and right ears. Figure fig. 3.3 shows how these two properties enable directional hearing. Firstly, the head itself creates a physical barrier that attenuates sound waves at one ear more-so that at the other ear. Secondly, the difference in distance between the sound source and each ear will create a time delay between the received signals. Over time we learn to associate patterns in ITD/ILD with the physical location of the sound source.

The relative importance of ILD and ITD differ depending on the frequency of the incoming sound. The reasons for this are twofold. Firstly, a person's inter-aural distance (the physical distance between their ears) must be longer than the wavelength of the incoming sound wave in order for them to detect the phase difference between wave peaks. In this case, it is often referred to as Interaural Phase Difference (IPD), which accounts for the centre frequency of the critical band. Secondly the cilia that line the basilar membrane inside the cochlea can only detect the fine structure of a sound wave up until roughly 1.6 kHz [23]. Lindemann gives a rule of thumb that time differences of less than 7.5 ms are heard as a single auditory event (for which direction could be inferred), and differences above 21 ms are heard as uncorrelated events [23]. Hence, at 1.6 kHz and below ITD is more important, whereas above 1.6 kHz ILD is more important. Furthermore, Blauert's subsequent work found firstly that both ILD and ITD are relevant within the frequency band surrounding 1.6 kHz, and secondly that the angle of incidence of the sound wave plays a role in how the cues are perceived [24]. Despite this, Lindemann's rules of thumb are still broadly accepted, even being encoded into international standards such as [25]. As an example, fig. 3.4 shows two recordings (a high-frequency hi-hat and a low-frequency kick drum) that originate from the right-hand side of a listener. The high-frequency hi-hat has a large ILD and am-

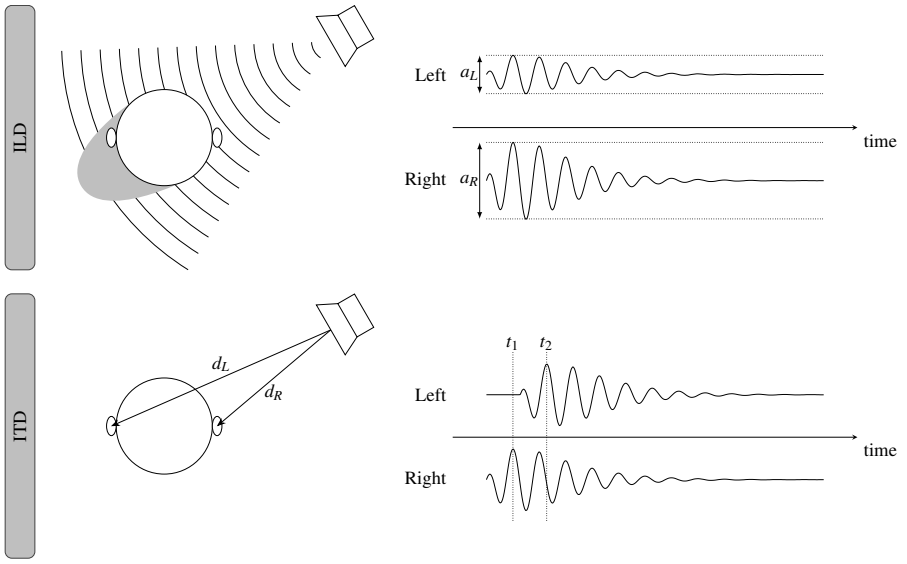


Figure 3.3. Illustration of how humans perceive the direction of an incoming sound. The physical barrier of the head causes an Interaural Level Difference (ILD) between sound waves received at the two ears ($a_R - a_L$). The difference in distance from the source to the ears ($d_R - d_L$) causes an Interaural Time Difference (ITD) between the two ear ($t_2 - t_1$).

biguous IPD, whereas the low-frequency kick drum has negligible ILD, but an interpretable IPD.

3.1.3 Loudness perception

While we have established that the ILD is a key factor of directional hearing, a more detailed understanding of the link between perceived loudness and sound wave intensity is necessary for musical applications.

We perceive the loudness of sound, not in terms of the absolute amplitude of the sound wave, but the *power* of the sound wave. Mathematically speaking, perceived loudness is proportional to the Root Mean Squared (RMS) Sound Pressure Level (SPL) raised to an exponent [26]. Observing that the calculation of RMS naturally involves taking the square of the original time function (see eqs. (3.1) and (3.2) for the continuous and discrete forms respectively), it follows that multiplication of a signal by $\sqrt{2}$ will result in double perceived loudness. Conversely, multiplying a signal by $\frac{1}{\sqrt{2}}$ will halve the perceived loudness.

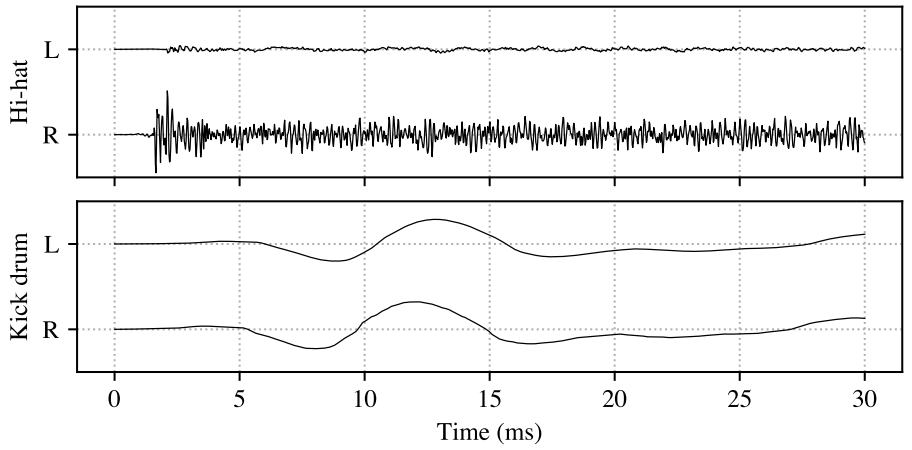


Figure 3.4. Example sounds coming from the right-hand side, recorded binaurally. High-frequency content (above) has a large ILD and ambiguous IPD, whereas low-frequency content (below) has negligible ILD, but an interpretable IPD.

$$f_{RMS}(t) = \sqrt{\frac{1}{T_2 - T_1} \int_{T_1}^{T_2} [f(t)^2] dt} \tag{3.1}$$

$$f_{RMS}[n] = \sqrt{\frac{1}{N} \sum_{n=0}^{N} f[n]^2} \tag{3.2}$$

This basic rule is relevant when two sounds are mixed in music production. It defines a precise, mathematical relationship between perceived loudness and the amplification applied by the production equipment or software.

3.2 Music production

The principles of our auditory perception have been exploited for creative use since the turn of the 20th century. Early stereophony came from the world of cinema, with sound technicians adding multiple speakers to enable spatial effects[1]. Eventually migrating into the music industry, stereophonic recording became the norm in the 1950s [5, 27, 28]. While both cinema and music practitioners have long known that using two speakers for reproduction enabled spatial effects, knowledge of how to *control* those effects came with the psychoacoustic understanding detailed in section 3.1.

[1] A typical use case was to match the origin of a voice with the actor's location on the screen.

3.2.1 Panning

The concepts of directional hearing are applied in music production to make sound sources appear as though they originate from a specific direction or position, often referred to as the 'phantom position'. In recording studio vernacular, this is call 'panning'. In accordance with the two signal differences a that humans are able to detect (ILD and ITD), there are also two basic types of panning – amplitude panning and time panning. By way of example, fig. 3.5 shows how the phantom position can be moved anywhere in between the speakers that are reproducing the sound by applying different amplitude gains to the left and right channel of a stereo recording.

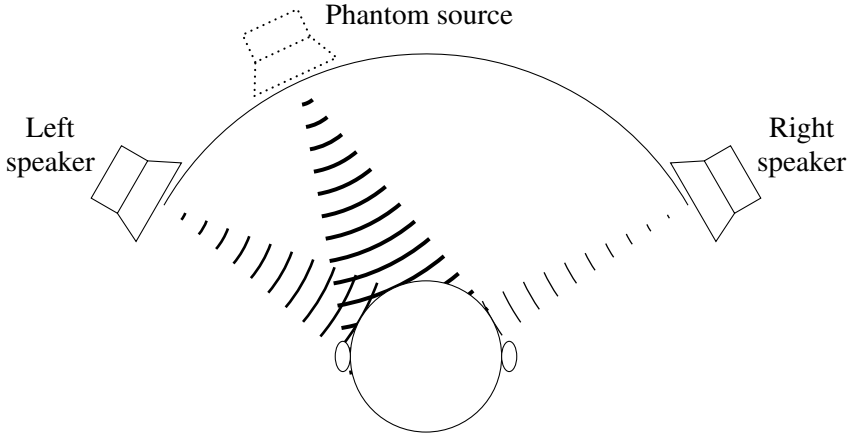


Figure 3.5. Perception of a phantom sound source created by amplitude panning. The sound wave from the left speaker has higher amplitude than that of the right speaker, thereby creating a phantom source which appears to come from the left.

Time panning (otherwise known as the 'Haas effect') exploits the phenomenon that the first ear to hear a sound generally dictates the direction that it appeared to come from. The process is to simply delay the signal in one channel by a time value within the 'Haas window' (typically less than 40ms). The un-delayed channel will then dictate the phantom direction of the original source. In music production it is used both for panning and to add spaciousness to a recording [27, 29].

Another frequently used technique is 'binaural panning', which assigns combinations of amplitude and time difference to each stereo channel. This is typically achieved by using a Head-Related Impulse Response (HRIR). An HRIR is a pair of impulse responses recorded with microphones placed in the ears of either a real person or a dummy head. To obtain an HRIR, a known excitation signal is played from a specific position in a room and recorded with said binaural microphones (see fig. 3.6). The HRIR can then be convolved with any other monophonic recording to make it appear as though the recording came from the position of the excitation source. An HRIR can only be

used to pan a sound to a single position, so it is typical to record many from different positions [30].

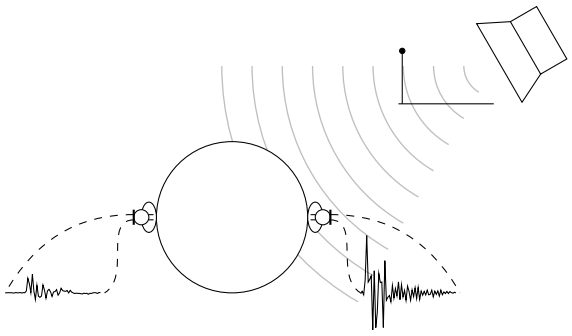


Figure 3.6. Illustration of a binaural HRIR being recorded. An impulse is played back through the loudspeaker and recorded at microphones placed in each ear. The head acts a filter, altering the left and right microphone signals differently.

In traditional recording studio setups, the recording engineer or producer will use one or more microphones to capture an original sound source. These singular sound sources are commonly called 'stems'. During the production process, the stems may be panned to create a sense of space in the final mix. In the case that the stem was recorded using two microphones, the stem would be called a 'stereo stem', and the term 'panning' is usually replaced by 'balance', and only amplitude panning (as opposed to time or binaural panning) would be applied.

Amplitude panning is the most common panning technique in traditional music production [31]. It is associated with so-called 'panning laws', which govern how the amplitude is changed in each channel according the desired pan position [27]. One example is the 'energy-preserving pan law' (sometimes called the '-3dB pan law') in which the squared sum of the channel gains always equals 1. The gains of the left and right channel are calculated by $g_L = \cos(\frac{\pi}{2}a)$ and $g_R = \sin(\frac{\pi}{2}a)$ respectively where a is the panning position in the range $[0,1]$. Figure 3.7 shows how the channel gains relate to the panning position. The total perceived loudness will be equal at all panning positions.

Panning techniques are applied to single stems only. A typical recording session will contain multiple stems, each with its own panning.

3.2.2 Stereo image

The term 'stereo image' is used to describe the overall distribution of sound across the horizontal plane after panning and any other production techniques have been applied on all stems. One may also use terms like 'stereo field', 'soundstage', or 'panorama', which all refer to the same thing. They are usually described in qualitative terms such as 'big', 'wide', and 'ambient' rather

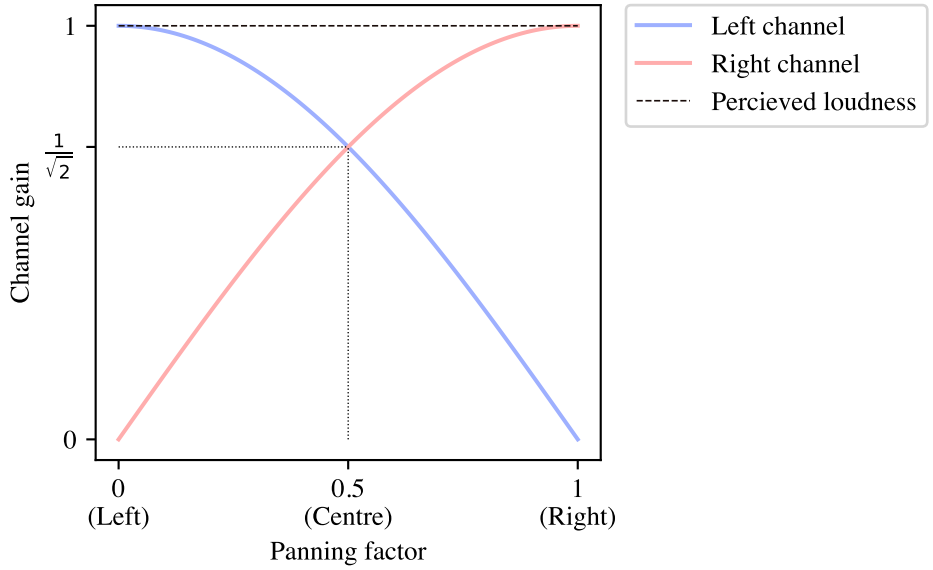


Figure 3.7. Channel gains for equal power panning. When both channel gains are $\frac{1}{\sqrt{2}}$, the sound will be centre-panned. The perceived loudness is equal at all panning positions.

than quantitative measures [32]. When mixing a musical recording, a producer will pan individual stems to their desired position within the stereo image. Those stems which are recorded in stereo in fact have their own stereo image, which can also be controlled by the producer. The combination of all stems into a final stereo mix, constitutes the overall stereo image.

Figure 3.8 shows a simple example stereo image, as perceived by the producer. It contains two mono stems – vocals and violin, and one stereo stem – piano. There are some common tropes seen throughout many productions, some of which are shown in the diagram. Firstly, vocals are recorded in mono and panned to the centre. Secondly, since the piano is panned slightly left, the violin is panned roughly the same amount to the right in order to achieve a balanced mix. Thirdly, all stems are placed in the stereo image in a way that emulates a natural performance on a stage, and to avoid separate instruments masking each other. Note that while these practices are commonplace, they are purely an artistic choice on behalf of the producer [32]; Recordings breaking the trend are not difficult to find.

Describing the stereo image of a recording has its roots in the acoustics domain, in which typical use cases are the analysis of concert halls and music reproduction systems. In these cases, the previously mentioned spatial cues (ILD, ITD, and IACC) typically form the basis of the analysis. Berg and Rumsey give a long list of typical perceptual measures that can be derived from the low-level cues [33], although later literature has focused on the measures of

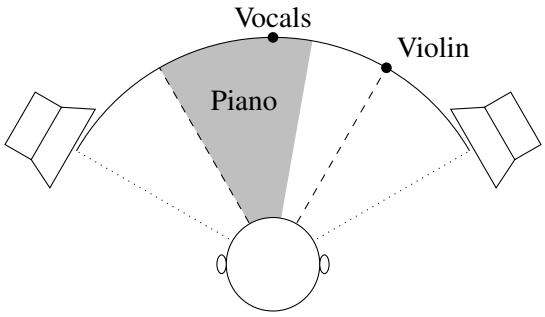


Figure 3.8. Example stereo image created by a music producer. Mono stems are shown as black dots and a stereo stem as a grey fill. Dashed lines represent the extent of the stereo image. Dotted lines represent the maximum boundaries of the possible stereo image.

Apparent Source Width (ASW) and Listener Envelopment (LEV). However, as stated by Sarroff and Bello, the task of describing the spatial aspects of recorded music is inherently different to that of acoustical signals since said recordings may exhibit spatial aspects that cannot occur in physical acoustic space [34].

3.2.3 Production equipment and software

To create a stereo image, a music producer will use the equipment available in a typical recording studio. Before the advent of digital recording, said equipment usually comprised a mixing console where each stem was assigned to one 'track'. Each track had individual controls for (amongst others) level and panning that were directly linked to the internal electronics. The producers might also use the term 'fader' for the level control and 'pan-pot' for the panning control[2]. All tracks would be mixed by the console and the combined output fed to stereo speakers for playback. In a digital recording studio, the equipment is largely the same except that the mixing console will be connected to a computer that controls the session and stores the recorded audio. fig. 3.9 shows photographs of a small recording studio where the faders and pan-pots for each track are visible. The music producer will manipulate the physical controls for each track to create a stereo mix.

In modern recording scenarios a music producer typically uses a DAW - graphical software that allows them to record and edit audio. There are many DAWs available, but one such example (Reaper) is shown in fig. 3.10a. In particular, it shows a single recorded audio track which has its 'Pan' control automated. It also contains a single effects track which applies a stereo reverb effect to the recorded audio. This effects track has its 'dry/wet' mix parameter

[2]The term 'pan-pot' is a shortening of 'panoramic potentiometer'.



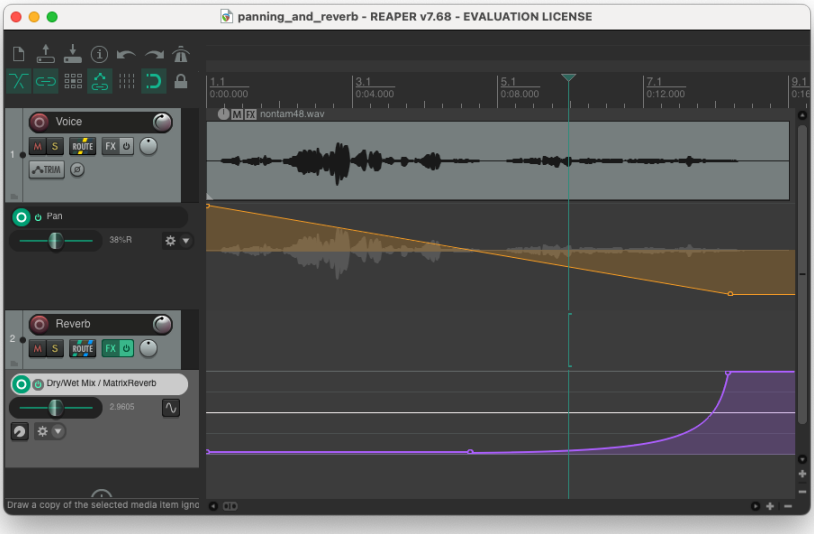
Figure 3.9. Photographs of a small recording studio*. The left photo shows all studio equipment, while the right photo shows a close-up of the mixing console. Faders are highlighted pink, pan-pots are highlighted blue, stereo speakers are highlighted green, and the computer running a DAW is highlighted red. One track is highlighted yellow.

*Original images by cottonbro studio are licensed under Pexels. Annotations have been added.

automated, which controls the ratio of non-processed to processed sound. In a more complete recording session, the DAW would contain many more audio and effects tracks, each with their own pan control. FiXme: Do we need to show the effects track?

The process of creating a stereo image is largely a subjective one. The producer pans each track individually by either physically turning the pan-pots or adjusting a digital panning control, and judges the stereo image purely by listening. In a large DAW session containing many tracks, the producer will make artistic choices about where in the stereo field to pan each track. While there are some typical practices in the industry, it must again be emphasized that panning and creation of stereo field more generally is an artistic process [27].

In some cases the producer consults monitoring equipment such as a level meter or goniometer to fine tune their mix [12, 37]. They also look for potential quality issues that can occur when the stereophonic recordings are mixed down into a monophonic signal for reproduction, which have been documented well before the advent of digital recording [38]. In more modern production, DAW plugins are generally used to visualize and assess the stereo qualities of the final mix [12, 39]. Figure 3.10b shows one example - the ToneBoosters Goniometer plugin [36]. It is designed to replicate an analogue goniometer, which, taking two voltage signals as input, displays them on a 2D axis such that a music producer might be able to discern the correlation between the signals.



(a) Recording project in the Reaper DAW [35]



(b) ToneBoosters Goniometer plugin [36]

Figure 3.10. Screenshots of example music production tools

3.2.4 Audio representations

Audio representations are standard paradigms that an audio recording can follow. If we consider an audio recording as a set of concurrent time signals, the representation defines how each signal in the set should be interpreted. They are usually grouped into 'channel-based', 'scene-based', and 'object-based'. In channel-based audio, each signal represents the exact waveform that should be played back by a single loudspeaker. In scene-based and object-based audio, each signal represents one part of a larger scene, and rendering equipment is responsible for translating that scene to individual speakers [40]. While the focus of this thesis in on stereo audio, it is important to compare stereo to the wider audio landscape to understand why it is so heavily used for recorded music.

Monophonic

Monophonic recordings contain a single channel to be played back on one speaker. They are only able to convey a small amount of spatial information; reverberation is the only way to infer a sense of space. Archival/historical recordings will be recorded in mono, as well as a majority of raw studio stems and most speech/telephony recordings.

Stereophonic

Stereo recordings contain two channels: left and right. Compared to mono, stereo has the ability to convey the width and location of a sound 2D space. For music this sense of space is especially important since music signals tend to be highly dense and complex. Stereo mixing therefore allows music producers to add clarity and balance to their recordings. This marked increase in quality drove stereo's commercial adoption in the 1950s, and it has been the dominant format for recorded music since then [5].

Binaural

Binaural can be considered a subset of stereo. Each channel conveys the precise sound wave as heard at the left and right ears. It is typically used when rendering spatial audio[3] for playback on headphones, and in Virtual Reality (VR) / Augmented Reality (AR) scenarios.

Multichannel

Multichannel is a continuation of stereo which adds speakers at more positions. While it could contain any number of channels, there are some standards that are referred to by their channel count and layout, e.g. 5.1 (5 listener-plane channels, 1 Low Frequency Effects (LFE) channel) or 22.2 (10 listener-plane,

[3]The exact definition of 'spatial audio' is somewhat contested in the literature but here I assume Cobos et al.'s definition: *'audio that contains spatial information'* [41].

2 LFE, 9 overhead, 3 low) [40]. Compared to stereo, it has the ability to convey width, depth, and – depending on the arrangement – height, creating a higher sense of immersion. This ability sees it being used mainly for film and TV soundtracks where the emotional impact of single sound events can be critical. That said, the high setup cost and difficulty positioning extra speakers has historically discouraged its wider adoption [42].

Ambisonics
Ambisonics is a scene-based audio representation where each channel is one directional component of a 3D physical sound field. Recordings are referred to by their 'order', which implies both the number of different directions and the number of channels. The number of required channels increases quadratically with the order according to the formula $C = (O+1)^2$ where C is the number of channels and O is the order. Ambisonics typically excels at representing diffuse sounds, but struggle to represent point sources unless the order is very high. Thus, ambisonic recordings are only used for very specific types of music, such as orchestral performances in large concert halls where accurate recreation of the ambience is more important than individual instruments. Even in those cases, an ambisonic recording will generally be rendered to binaural for mass distribution [43].

Object-based audio
In object-based audio, the entire audio scene is composed of individual audio objects accompanied by metadata specifying their location and movement. Rendering equipment is then responsible for translating those objects to speaker channels for playback. It is a relatively new format and while its main use is for cinematic sound, the music industry has recently begun to adopt it. In contrast to ambisonic recordings, object-based audio is distributed in its original form, allowing the same set of signals to be rendered in multiple ways.

Overall, stereo recordings are able to deliver high enough quality for the majority of music listeners. The newer formats, while gathering interest in pockets of the music community, have yet to receive wide-spread adoption from consumers. The result is that the mainstream distribution of music still happens in stereo.

3.2.5 Lossless and lossy encoding

Regardless of which audio representation is chosen, the resulting recording will most likely be encoded to a compressed format for distribution. There are two types of compression: 'lossless' and 'lossy'. A lossless format, when decoded, will produce bit-for-bit recreation of the input signal. Lossy formats on the other hand will not produce the exact same signal, but will discard

information that they consider irrelevant according to a psychoacoustic model. There are many lossy formats each suitable for different types of audio, but for the purposes of this thesis it is just pertinent that spatial information can be lost during lossy encoding.

3.3 Analysis of digital recordings

The output of a recording session is usually only the stereo mix. Stems and DAW sessions are generally kept private, with precious few being available under licensing terms that are conducive to academic research [44]. Thus, we assume that all analysis can only be performed on the final stereo mixes that are distributed for playback. The most common distribution format is a digital audio file, and there are some fundamental analysis techniques that can be applied to a digital audio signal.

3.3.1 Digital audio signals

When a sound wave interacts with a microphone, it generates an electrical signal in continuous time. To convert that analogue signal into a digital one, it is sampled at regular intervals. The sampling frequency (f_s) is related to the sampling interval (T_s) by the relationship $T_s = \frac{1}{f_s}$.

Shannon's sampling theorem states that a signal can be perfectly recreated when sampled at twice the highest frequency contained in the signal. Conversely, the highest frequency that can be represented by sampling a signal at a given f_s is half of that frequency. This limit is called the 'Nyquist frequency' and is calculated by $f_N = \frac{f_s}{2}$ [45]. This is illustrated by Figure 3.11, which shows the sampling of three waveforms. The lower plot shows a signal at the Nyquist frequency (f_N), which can be perfectly reconstructed from the digital samples. The upper two waveforms have frequencies equidistant but in opposite directions from the Nyquist frequency ($f_N \pm f_\delta$). The digital samples shown in circles could represent either of the two waveforms. In practice, when the analogue waveform is reconstructed, the lower frequency will always take precedence.

A typical f_s for music is 44.1 kHz, with a resulting f_N of 22.05 kHz. The prevalence of this particular number is due to the popularity of Compact Discs (CDs) as a distribution medium[4]. While higher frequencies may be used in the recording and production process, an exceedingly large proportion of recorded music has been distributed at 44.1 kHz.

[4] 44.1 kHz was chosen by a consortium of CD manufacturing companies as a trade-off between technical requirements and the limits of human perception [46].

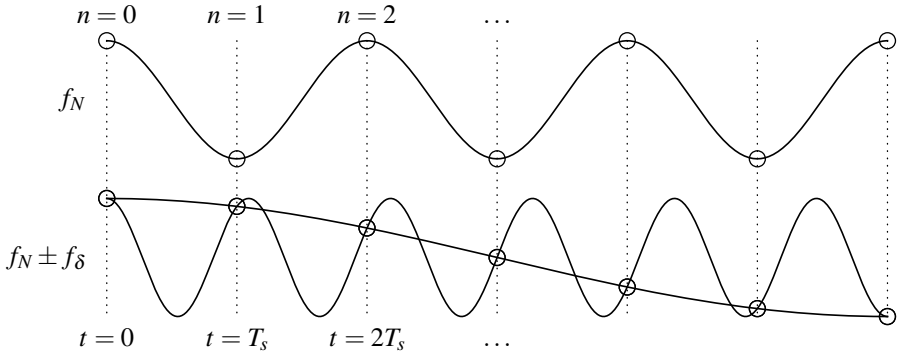


Figure 3.11. Illustration of the Nyquist Frequency in digital audio signals. Dotted vertical lines represent sampling points with a separation of T_s, and monotonically increasing index n. The upper plot shows a cosine wave at exactly f_N. The lower plot shows two cosine waves with frequencies equidistant above and below the Nyquist frequency (f_N). Sampled values are shown as circle marks on the plots. The two upper signals can both be represented by same digital samples.

3.3.2 Time-Frequency representation of audio

In order to perform many types of audio analysis, the audio signal can be converted from the time domain into the time-frequency (TF) domain. In perceptual acoustics research there are several common forms of TF representation, each with its own nuances. This section provides those details and Figure 3.14 shows a visual comparison of all spectrogram types.

Short-time Fourier Transform (STFT) spectrogram

The STFT spectrogram is one of the most popular TF represention across all audio processing work due to its computational efficiency. The fundamental principle is that a time-domain signal is split into short (1 to 100ms) overlapping chunks, and the Fast Fourier Transform (FFT) is performed on each chunk, thereby converting that chunk into the frequency domain. By arranging subsequent frames along a time axis it is possible to represent the original signal as a 2D image known as a spectrogram. When the time-domain signal is entirely real, (which is the case for monophonic audio signals), the result of the FFT will be a complex signal which is symmetrical around the Nyquist frequency and the upper half is therefore dropped. Furthermore, the complex spectrum is usually represented in magnitude/phase form to better match the human auditory response.

A fundamental property of the FFT is that it assumes that the signal repeats outside the time frame that is being analysed. Since this is generally not the case for audio signals, it is standard practice to apply a time-domain windowing function to each frame so that large transients do not occur at the

boundaries. Serveral standard windowing functions exist, each with their own nuances, but for most audio applications the 'Hann' window can be used [47].

The nature of the spectrogram is affected by two parameters: the frame size and hop size. 'Frame size' refers to the number of samples in each analysis frame. To obtain maximal computational efficiency this must be a power of 2. 'Hop size' refers to the number of samples between the start of subsequent frames and is sometimes expressed as a percentage of the frame size. The frame size dictates the resolution of the spectrogram's frequency axis since the frequency bins will be spaced evenly between 0 Hz and the Nyquist frequency. Importantly, a longer frame size will result in lower time accuracy within each frame; sharp transients in the time domain will appear as wideband energy in the frequency domain, but information about exactly where the transient occurred within the frame is lost. To counteract this inaccuracy, a smaller hop size can be used so that transients might be detected as sharp changes between STFT frames. One major drawback of the STFT spectrogram is that the linearly-spaced frequency bins do not align well with the human perception of frequency, which is logarithmic. The STFT therefore contains redundant information in higher frequencies. Several common alternatives exist to account for this.

Mel spectrogram

The 'Mel scale' is a quasi-logarithmic frequency scale derived from perceptual experiments. Several definitions exist in the literature [48], but popular implementations like that of librosa [49] and Auditory Modelling Toolbox (AMT) [50] tend to underpin most audio experimentation.

A 'Mel filter bank' is a set of band-pass filters that are spaced equally on the Mel scale. Practically, they are implemented as sets of weights that are applied to an STFT spectrum with the relationship $M_n = \sum_{k=0}^{N} S_k W_{k,n}$ where M_n is the nth Mel coefficient, S_k is the kth STFT bin, N is the number of STFT bins, and $W_{k,n}$ is the kth weight of the nth Mel filter. An example Mel filter bank with 8 filters is shown in fig. 3.12. While 8 filters are shown for demonstration purposes, audio analysis typically requires over 100.

When this filterbank is applied to each time step of an STFT spectrogram, the result is a 'Mel spectrogram', an example of which can be seen in fig. 3.14. Observe that the higher frequencies have been condensed into fewer frequency bins compared to the STFT spectrogram.

Cochleagram

As an alternative to the two spectrogram representations, a cochleagram aims to make a TF representation that models the acoustic response of the human cochlea more closely. To that end, all processing is performed in the time domain and is sometimes referred to as a 'peripheral ear model'. Several examples appear in the literature [52–54]. In the classical method [52] the original signal is first processed by a bank of gammatone filters with their

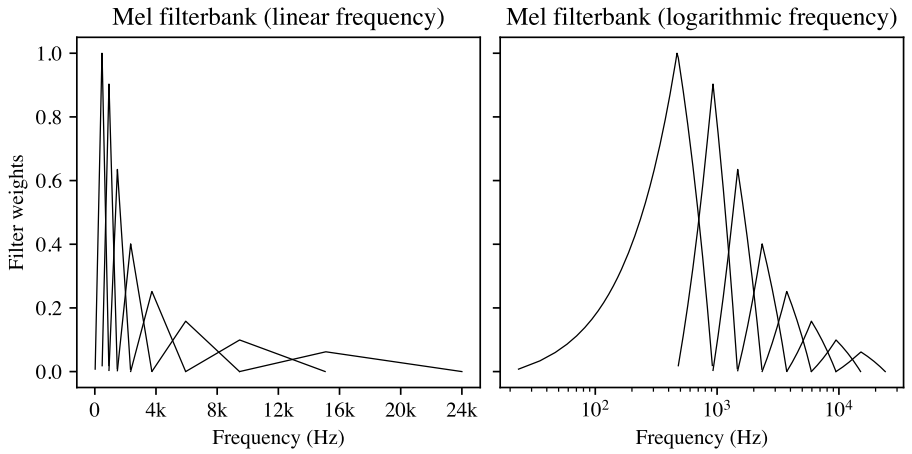


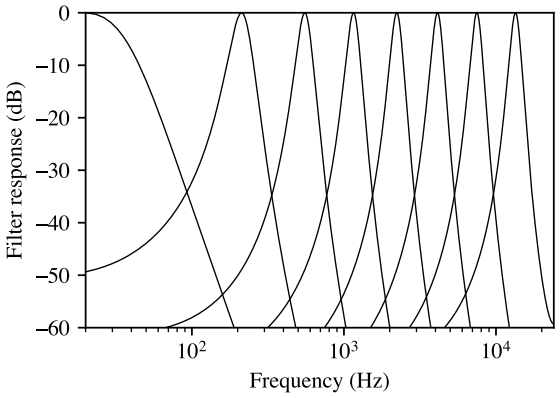
Figure 3.12. Example Mel filter bank containing 8 filters, shown with linear (left) and logarithmic (right) frequency scales. Note that the bandwidth of each filter is roughly equal in logarithmic space. Since each filter is designed to give a weighted average of linearly spaced FFT bins, the amplitudes are scaled so that the sum of the weights in each band is equal. The filters are calculated using librosa [51] using a sample rate of 48 000 Hz and an FFT size of 4096. Filter weights equal to zero are removed for clarity.

centre frequencies placed according to a perceptual scale (such as the Equivalent Rectangular Bandwidth (ERB) scale [55]), as shown in fig. 3.13a. As with Mel filters, several definitions of gammatone filters exist in the literature, but the popular implementations (e.g. scipy [56] and AMT [50]) preside.

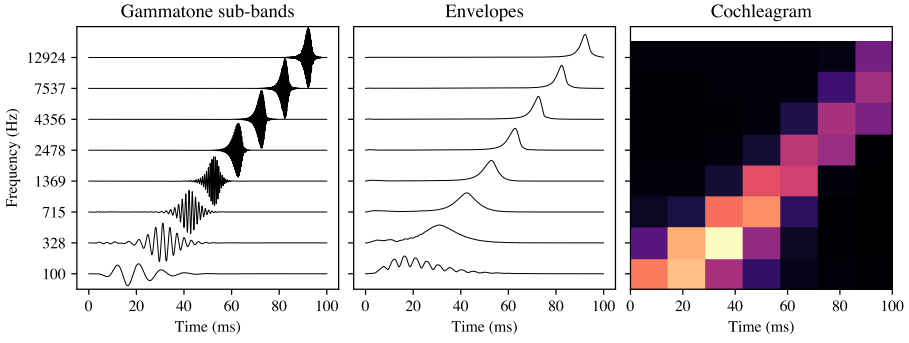
Figure 3.13b shows the intermediate steps to generate a cochleagram. Each sub-band signal is processed by a half-wave rectifier, square root, and low-pass filter to mimic the excitation patterns of the cilia and basilar membrane. The result is a signal representing the envelope of each sub-band. Those envelopes are then segmented into overlapping time chunks. The amplitude of each chunk is then calculated as the mean of all its values, forming one pixel in the cochleagram.

3.3.3 Machine learning

Any data-driven audio research, particularly that using vast datasets, will naturally involve the use of ML. Broadly speaking, ML techniques can be classified as either supervised or unsupervised. When training a supervised ML model, the training dataset needs to include ground-truth labels such that the network can objectively discern via a loss function whether the predictions it is making are correct. Unsupervised ML on the other hand does not require any labels. Instead, it seeks to form a representation of the underlying dataset via techniques such as clustering or dimensionality reduction, thereby learning



(a) Example gammatone filterbank with individual filters spaced equally on the ERB scale. The filters are Infinite Impulse Response (IIR) and are applied to an audio signal in the time domain.



(b) Steps of an example decomposition from time-domain audio to a cochleagram. The audio is a frequency sweep from 100 to 20000 Hz. First it is processed by a gammatone filterbank, generating 8 sub-band signals which are spaced equally on the ERB scale. Each sub-band has its envelope extracted. The envelope is then split into overlapping time chunks, which are averaged to form the cochleagram.

Figure 3.13. Gammatone filterbank

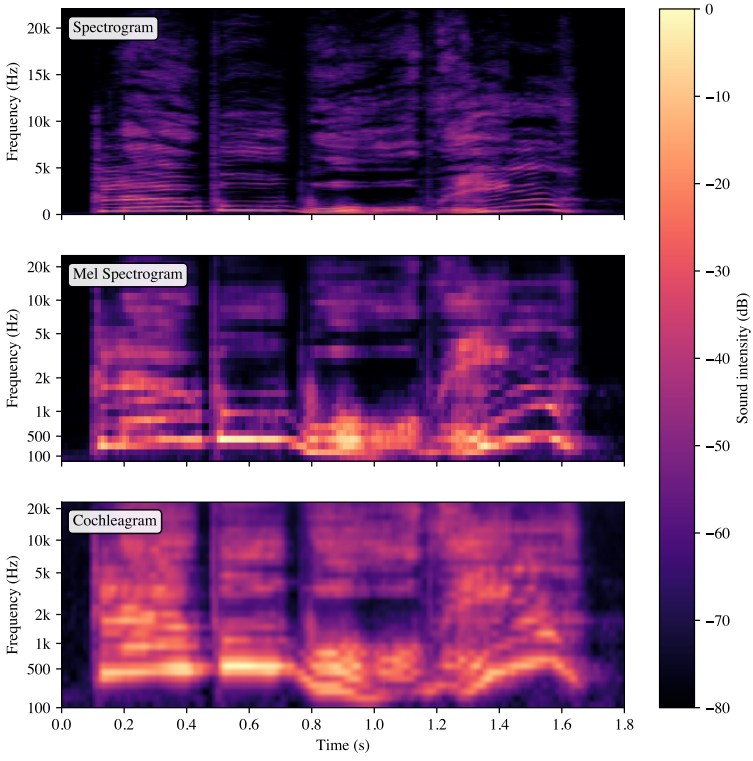


Figure 3.14. An example audio signal displayed as STFT spectrogram (above), Mel spectrogram (mid), and cochleagram (below). The spectrogram frequency bands are spaced linearly between 0Hz and the Nyquist frequency, which is not coherent with human perception. The Mel spectrogram is created from the original spectrogram based on linear combinations of its frequency bands and retains more detail in lower frequencies than higher frequencies. The Cochleagram is created directly from time domain audio by mimicking the physical response of the human ear. In all cases the pixel colour represent the sound intensity of the given TF bin in decibels.

the structures and patterns of the data while retaining as much key information as possible.

Striking a middle ground between the two is 'self-supervised' learning, which addresses the issue of creating objective ground-truths from unlabelled data. Instead of relying on external labels, self-supervised learning generates pseudo-labels from the dataset itself, and then uses those as the input to a 'pretext task', which is tackled in much the same way as a supervised task.

As an example we can consider the current largest dataset of recorded music: FMA [10]. The dataset contains the stereo audio of approximately 100 000 tracks, along with metadata such as genre tags. An unsupervised task might involve learning a high-order representation of the raw spatial cues present in the stereo signals. A supervised task might involve predicting genre using the audio signal as input. A self-supervised task might involve masking off a short period of audio and filling it back in using the surrounding audio as context.

In relation to the overarching RQ in section 2.1, we assume that a music producer's DAW session is the only source of objective ground-truth data regarding stereo field. Furthermore, we assume that DAW sessions are not publicly available, as was found in [57, 58]; The only public-facing output of a recording session is the mixdown. It follows that either unsupervised or self-supervised learning is necessary to draw any conclusions to the RQs.

4. Related works

In this chapter I break down the details of existing algorithms for stereo field analysis, as well as demonstrating those techniques on example audio recordings. To provide additional context, I also present several high-level initiatives within the audio industry, showing how stereo field analysis has been used up until the present day. Throughout the chapter I use 3 recordings as examples. They all come from the FMA dataset [10], and each one presents a different use of stereo image. In the text they are referred to simply as track 'A', 'B', and 'C'. Full details for all tracks can be found in appendix A.

4.1 A brief history of stereo field in industry

Research into the stereo field of recorded music specifically (as opposed to generic spatial properties of audio signals) has its roots in the early 2000s. At the time, stereo recording was a well-established norm and audio distribution was handled by the early lossy encoding formats such as MPEG layer 3 (MP3) and Advanced Audio Coding (AAC) [25, 59]. Analysis and description of stereo field was used to increase the coding efficiency of these codecs, for example by e.g. Parametric Stereo (PS) coding [60–62]. In parallel, new audio encoding schemes such as Binaural Cue Coding (BCC) (later ratified as 'MPEG Surround' [63]) were developed based on the work of Baumgarte and Faller [64].

The standardization of multi-channel audio presented an opportunity: new reproduction equipment needed to remain compatible with the existing back-catalogue of stereo recordings, and vendors could compete on sound quality. There was therefore a need to invent algorithms that upmixed stereo audio to multi-channel. Analysis of the stereo field, source separation, and primary/ambient extraction were seen as an effective way to achieve this [65]. In fact, such research is still ongoing today [66].

In the prevailing years, object-based audio has been developed and standardized as e.g. Dolby Atmos [67] and MPEG-H [68], and the same early research into stereo field analysis is still relevant to these new formats for all the same reasons. Nowadays, the same tasks are also being tackled using ML, and many of the new approaches out-perform their DSP-based predecessors [**findsomething**].

With access to these efficient distribution formats, large digital music collections emerged and brought new problems to solve: collectors needed ways

organize and categorize their music. Stereo field analysis was just one of the mechanisms used to facilitate this, paving the way for research into e.g. Automatic Genre Recognition (AGR), MIR, and music recommendation [10, 69, 70].

4.2 Spatial cues for stereo image

A 'spatial cue' is a low-level analysis of spatial information in a short (1-100ms) audio snippet. While those mentioned in section 3.1.2 aim to analyse binaural sound as perceived by humans, there are several examples of spatial cues tailored specifically to the analysis of recorded music.

The 'Panning Index (PI)' and 'ambience index' specified by Avendano and Jean-Marc [71] are loosely analogous to ILD and IACC in the acoustics domain. To estimate the PI, a signal model is assumed whereby a musical source is panned within the mix using the 'energy preserving pan law', in which amplitude differences are applied to each channel such that the sum of the squared amplitudes equals 1 (see section 3.2.2). While this panning technique does not affect the phase of either channel, existing phase differences between channels are measured using the 'ambience index', which is defined such that it is low when the stereo channels are correlated (indicating a direct sound source), and high when channels are uncorrelated (indicating an ambient or reverberant sound source).

Presti provides a re-contextualization of the panning and ambience indices in his work on Bivariate Mixture Spaces (BMSs) [72, 73]. He demonstrates that it can be used to analyse the panning position of individual TF bins, plus the clarity (or trustworthyness) of that measurement. Thus, it is useful for tasks such as primary/ambient decomposition, stereo field manipulation, and visualization.

Barry and Lawlor present another seminal technique in their paper defining the Azimuth Discrimination and Resynthesis (ADRess) algorithm [74]. The fundamental principle is to estimate the panning position of frequency bins in a stereo spectrogram by scaling one channel, inverting it, and mixing the channels back to mono. The resulting phase cancellation will determine the scaling factor that was applied to the original audio source during amplitude panning, and hence will show where the source was panned in the stereo field. To demonstrate, fig. 4.2 shows an example Frequency-Azimuth (FA) plane for a combination of panned sine waves. While this was technique was originally intended to be used for audio source separation, it has been repurposed many times since.

Importantly, all the previously mentioned analyses are performed in the TF domain and result in one value per TF bin. Their use cases typically involve reconstruction or editing of the original audio source as opposed to *description* of it since they inherently entangle semantic content with spatial information.

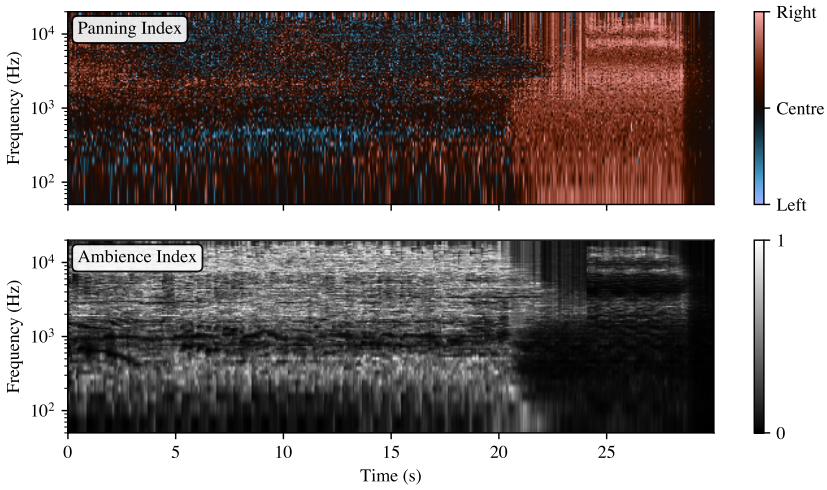


Figure 4.1. Demonstration of 'panning index' and 'ambience index' on Track C. Several panned sources are visible as red or blue bands in the 'panning index', and at roughly 20s most sources fade out leaving a single one on the right. The 'ambience index' plot shows that the track starts off with no single dominant direct sound source. At 20s when most sources fade out, the 'ambience index' drops to 0 across the lower half of the spectrum, indicating that the remaining source is direct. In the upper half of the spectrum, the source remains slightly ambient.

They have, however, become the basis of other algorithms that are designed specifically for description.

4.3 Stereo image description

Stereo image description refers to the generation of a numerical vector that represents the stereo image in an arbitrary feature space. Recordings with similar stereo image should be close within the space, although the distance metric is application-specific. A typical technique is to construct the vector from statistical summaries of low-level spatial cues. Throughout this section, these vectors are referred to as SIPs, and examples of each one are given on the three example tracks.

Tzanetakis et al. provide the earliest example of a technique specifically designed to interpret the use of panning in recorded music: Stereo Panning Features (SPF) [6]. It targets amplitude panning only, using Avendano and Jean-Marc's 'panning index' as the basis. Additionally, it separates the frequency spectrum into 3 sub-bands – lows, mids, and highs – following the common practice of music producers. While not specifically stated by Tzanetakis et al., I interpret the vector to represent the amount and variance of 'fast panning' and 'slow panning' in each frequency band. Tzanetakis et al. later

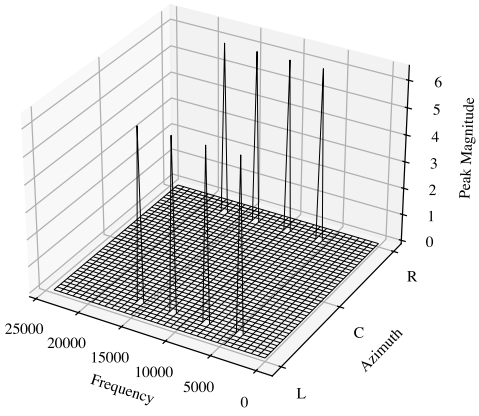


Figure 4.2. Frequency-Azimuth plane for a signal comprising 8 sinusoids at differing frequencies. Half are panned left and half are panned right.

expanded the features by adding an estimation of the number of individual sources in the recording [11]. The resulting SIP can be used to compare the recording and production styles of different music tracks, as shown in fig. 4.3.

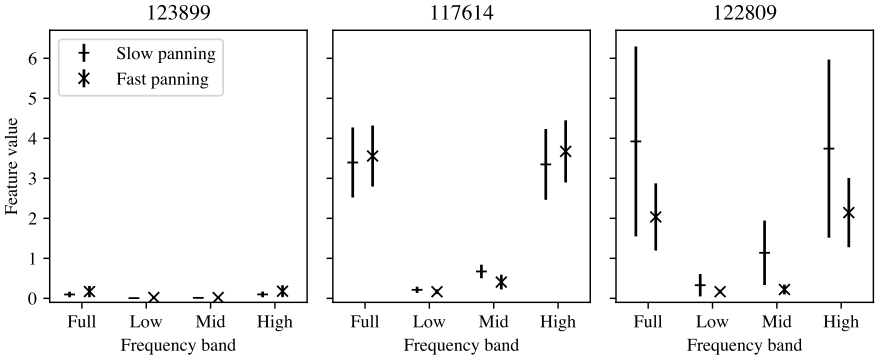


Figure 4.3. Stereo Panning Features for three tracks from the FMA dataset [10]. All tracks clearly show different panning and production styles.

While SPF shows production style at a high level, other descriptors contain more detail. Gomez et al. [13] and Sarroff and Bello [7] both present separate methods to form a Panning Histogram (PH), which displays the likelihood of a sound source being panned to a specific azimuth. Figure 4.4 shows examples of both methods for the same three example tracks. As Sarroff and Bello

state, the 'azimuth' value should not be interpreted as a true angular position relative to the listener, but rather as a loose placement of a sound source within the stereo field. Note that compared to SPF, PH visualizes the exact location of sound energy in the stereo field rather than just the 'amount of panning'. On the other hand, PH looses information about the *dynamics* of the panning, which SPF does not. Sarroff and Bello subsequently use the PH to calculate an estimate of stereo width [7], which is an important perceptual measure (see section 4.4).

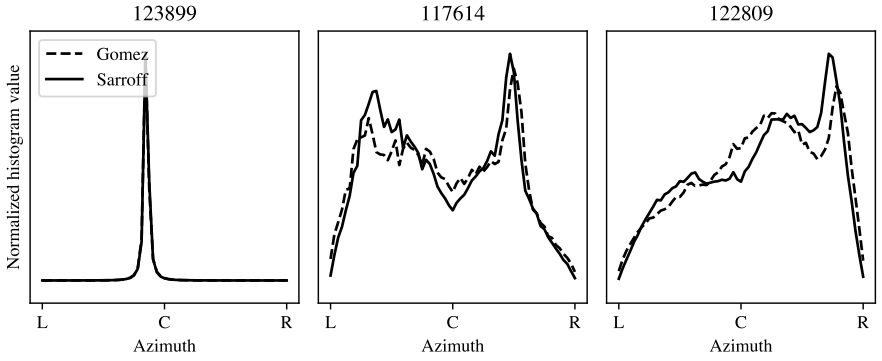


Figure 4.4. Panning Histograms for three tracks from the FMA dataset [10]. All tracks clearly show different shapes of stereo field. The two different algorithms (Gomez et al. [13] and Sarroff and Bello [7]) produce slightly different fine detail, but they agree on the general azimuthal distribution.

Gomez et al. present a continuation of their PH algorithm. They note that the histogram itself can be quite large, and that standard distance metrics may not correlate with perceptual differences. To avoid this, they propose converting the PH to a vector of 'Panning Coefficients (PC)' in a non-perceptual domain and truncating it. Equation (4.1) shows this conversion where $\boldsymbol{C}$ is the vector of coefficients, $\mathscr{F}^{-1}$ is the inverse FFT, L is the truncation length, and $\boldsymbol{HH}^{\leftarrow}$ is a concatenation of the panning histogram and its reverse. The efficacy of this technique remains under-tested, and my work in paper III showed that PC was unable to accurately represent PHs with sharp peaks.

$$\boldsymbol{C} = \left\{ \mathscr{F}^{-1} \left\{ \log(\boldsymbol{HH}^{\leftarrow}) \right\} \right\}_{1:L} \tag{4.1}$$

Tardieu et al. present another assessment of amplitude panning named Amplitude Stereo Features (ASF) [8]. Differing from the previously mentioned techniques, Tardieu et al.'s method focuses on providing measurements of the amount and variation of panning in many frequency bands. The basis of the algorithm is the ILD of a stereo cochleagram, as opposed to an STFT spectrogram. Tardieu et al. use 70 frequency bands equidistant on the ERB scale, each of which produce multiple features. Thus, the resulting SIP vector is an

order of magnitude larger than SPF or PH. As fig. 4.5 shows, ASF's large size allows for a very detailed assessment of panning across the frequency dimension, although it runs the risk of including noise and redundant information.

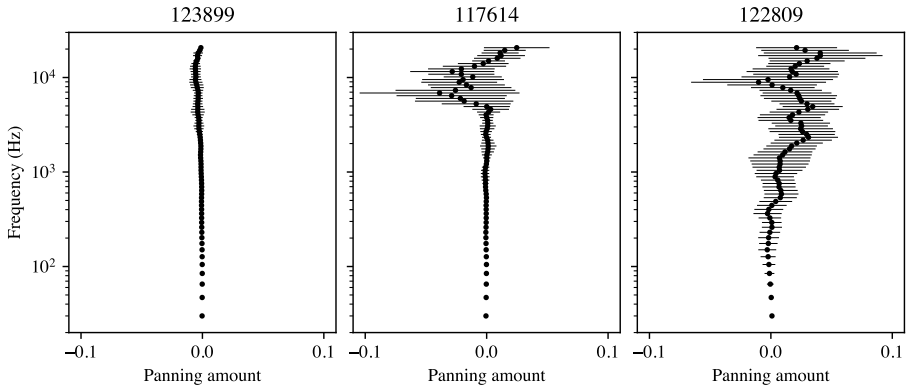
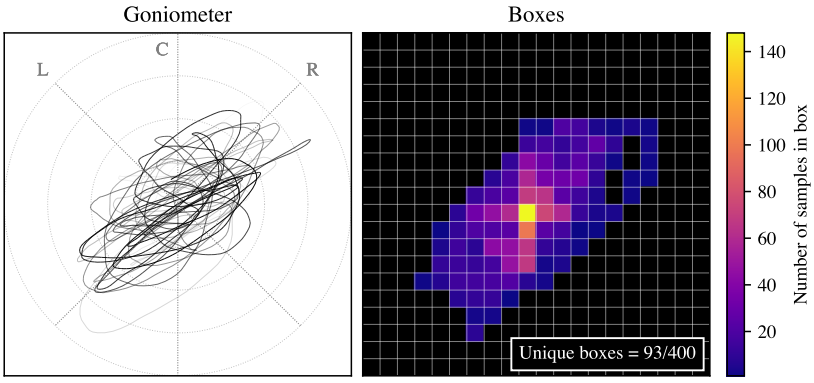


Figure 4.5. Visualization of ASF for three tracks from the FMA dataset [10]. For demonstration purposes, only a subset of the calculated features are shown. The amount and variation of panning is visible in fine detail across the frequency spectrum.

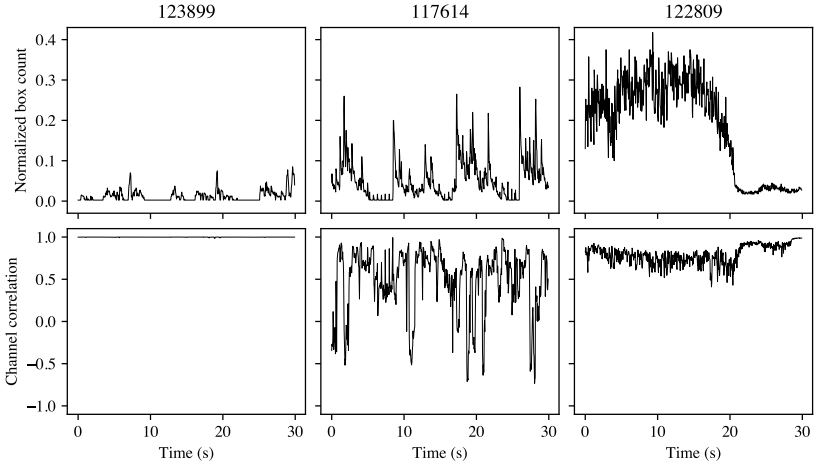
While all the aforementioned studies focus on descriptions of amplitude panning in stereo mixes, recording engineers can make use of phase-based techniques to add spatial feeling to their mix. Tardieu et al. also define a measurement for these named Phase Stereo Features (PSF) [8]. To enable this, they also define an STFT spectrogram derivative named Spectral Stereo Phase Spread (SSPS). FiXme: Unfortunately I cannot demonstrate it in this thesis as Tardieu et al.'s method produced spurious results.

Ziemer et al. argue the case in [12, 14, 37] that analyses based on a recording studio goniometer can aid in describing the stereo properties of music, particularly in differentiating between the artistic intentions of different recording sessions. The underlying logic is that since producers consult goniometers in their creative process, their readings must have inherent descriptive value. To perform the analyses, a virtual goniometer readout consisting of a lissajous figure and channel correlation measurement is constructed for non-overlapping time frames of an audio recording. The lissajous figure is quantized and reduced to a single number by counting the unique X/Y positions. Channel correlation is defined as the Pearson correlation coefficient [75] between the left and right channel signals. Figure 4.6 shows how an audio signal can be converted to goniometer features.

Härmä's work on the classification of stereo audio in [16, 17, 70] presents a slightly different take on the problem. He proposes a signal model where the stereo mix is formed from a combination of a single centre source, a significant amplitude-panned source, plus left and right uncorrelated sources. He subsequently presents a list of 8 pre-defined classes representing various com-



(a) Example goniometer reading for a 40ms segment of recorded audio. The left plot shows an original goniometer as the producer would see in their DAW. The right plot shows the boxes that the signal passes through as defined by Ziemer [12].



(b) Goniometer features as defined by Ziemer [12] for the three example tracks. Each track clearly shows a unique profile in how the features vary over time.

Figure 4.6. Example goniometer features

binations of those sources. Furthermore, he shows that stereo recordings can be classified into 6 out of 8 of those classes by estimating the variances of the underlying source signals. The classes are visualized as areas on the surface of a unit sphere, with each dimension representing the variance of the 3 source types, as shown in fig. 4.7. It should be noted that Härmä does not give a method for the classification of the final class – a mixture of all three source types – deeming it too complex to detect. In the context of stereo image description, Härmä's work is somewhat different from the others since the descriptions (i.e. the 8 classes) are pre-defined; The main innovation is the method to assign an audio signal to a class.

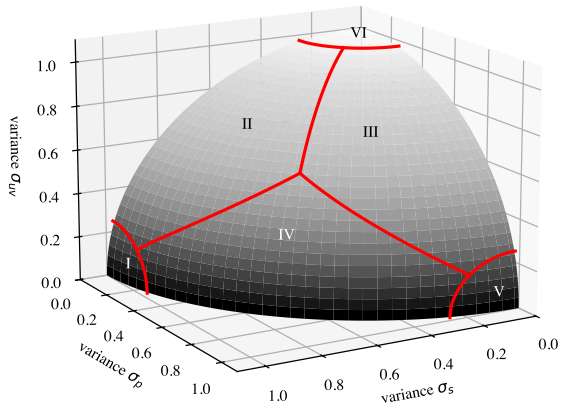


Figure 4.7. Visualization of 6 stereo mixture classes, recreated from [70]. The axes σ_s and σ_p represent the variances of the centre source and panned source, whereas σ_{uv} represents the covariance of the left and right uncorrelated sources. Classes are indicated as Roman numerals on the sphere's surface and separated by red dividing lines. Class 0 is silence and is hence trivial to detect. Class VII is cannot be detected by Härmä's method.

All the previously mentioned algorithms use classical DSP to create their spatial descriptors, but more recently there have been efforts to use DL to disentangle spatial information from audio content. Yang et al. presents one such example in [76]. While the original aim was to upmix stereo music to multichannel, a necessary component of the resulting neural network was a

latent space that represents the spatial information of the multichannel signal. A simplified architecture diagram of Yang et al.'s method is shown in fig. 4.8.

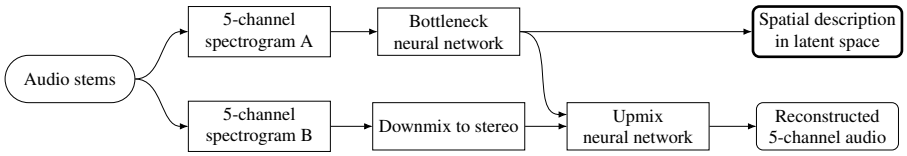


Figure 4.8. Architecture digram of Yang et al.'s neural network that learns a spatial representation of its audio training data. The latent vector (shown in a thick outline) can be used as a description of the spatial image. Spectrograms A and B use the same stems but different panning parameters. Recreated from [76].

DL has also been used outside the music domain to e.g. learn spatial relationships between individual microphones in an array [77]. While such tasks do not directly address the research questions in this thesis, they demonstrate that learning spatial features on large amounts of unlabelled audio data is possible given the appropriate methods and pretext task.

Neural generation of spatial audio also continues to be a hot topic up to the present day, with many attempts to generate high-quality binaural, ambisonic or multichannel audio from mono sources [78–84]. In doing so, the networks must infer spatial information from a source other than audio (typically an accompanying video or text prompt), and in the process will learn the spatial qualities of their audio training data.

4.4 Prediction of perceptual scalars

A perceptual scalar is a measurement that would usually be made via a perceptual listening test, e.g. ASW or LEV [85]. Those tests are known to be slow and costly [86]. Prediction of the perceptual scalars via objective means is therefore particularly useful to researchers as it allows for faster experimentation.

ASW is one of the more common values that has been addressed in the literature. The early work of Supper [87], Mason [88], and Dewhirst [89] detail the construction of 'binaural analysers' that are able to estimate ASW. Dewhirst compares several methods to estimate ASW given ILD, ITD, and IACC cues from binaural audio signals. The prediction results were promising, but mixed. Dewhirst even notes that the original values of ASW given by test participants were somewhat varied, concluding that the test itself was a difficult task.

Since these early experiments the analysis of ASW has remained a topic of interest, with Kasbach et al. [90], Mieth and Zölzer [91], and Antoniuk and Zieliński [92] all proposing ways to calculate it. It is not restricted to the binaural domain, with Sarroff and Bello [7], Wilson and Fazenda [93], and Mourgela et al. [94] also proposing formulas for generic stereo signals.

LEV on the other hand has prompted slightly more varied discussion over the years. An early study by Soulodre [95] compared several objective measures for it. The methodology was to fit a correlational model that predicted the output of perceptual test from objective audio measures. The objective measures were based on either the IACC or Lateral Energy Fraction (LEF) of the audio signal. Soulodre notes that the late-arriving sound ($>$ 80ms after the sound's onset) are most relevant to LEV, regardless of which objective measure is used. Sunish et al. [96], Power et al. [97], Jackson et al. [98], and Nowak and Klockgether [99] later expanded the same concept by fitting models on different objective audio measures.

Dewhirst et al. propose a different measure in [100] which involves calculating the area covered by the plurality of individual sound sources after estimating their directional localization. Although, by the authors' own admission, this is a crude measure and was not validated by perceptual tests.

Notably, the prediction of LEV appears to be used more often to assess spatial audio reproduction (sometimes rendered to binaural), rather than plain stereo. Power et al. [97] even include both mono and stereo reproduction in their experiment, concluding that their subjective ratings for LEV were considerably lower than all spatial formats. Sunish et al. [96] also state that LEV might not be cleanly separable from ASW.

Within the context of this thesis, the absolute accuracy of any prediction is not necessarily of primary importance. Rather, it matters that the value calculated for each recording in a corpus provides suitable comparisons to the other recordings. Mourgela et al.'s work on trends in music production [94] is one example of such a study. The study groups raw stereo width calculations into 4 buckets, thereby showing that finding recordings with similar stereo width is of more interest than the absolute value.

4.5 Spatial quality comparison

The SIPs described in sections 4.2 to 4.4 are often used to assess the spatial degradation of an audio signal after some sort of distortion has been applied. A typical method is to calculate the same spatial descriptor on both a reference and degraded signal, then calculate the difference in those descriptors. There is sometimes an additional aggregation step to reduce a multidimensional descriptor down to a scalar.

The types of distortion vary, although a major industrial use is in the development of lossy codecs. The work of You and Cheng [101], Wang et al. [102], Delgado and Herre [103], and Delgado and Herre [104] do exactly this. Figure 4.9 shows a typical workflow for the measurements of spatial distortion used in all these studies.

Spatial distortions other than lossy coding have also received attention in more recent years. For example, Barry et al. present a comparison of binaural

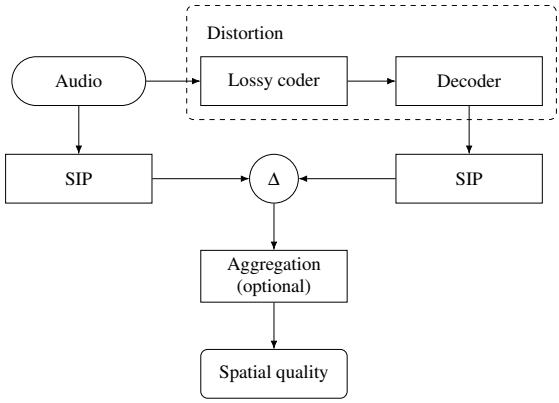


Figure 4.9. Example architecture to measure the spatial quality of a lossy audio codec. The two SIP blocks must use exactly the same method, although the choice of method is application-specific. The 'Distortion' block may be replaced when using the workflow for other tasks.

rendering and downmix algorithms (as well as lossy codecs), showing that minute differences in binaural cues can be easily detected using their methods [105]. Taking the opposite approach, Watcharasupat and Lerch demonstrates a new metric for assessing spatial quality and demonstrate it using several common, but artificial distortions [106].

In all the above studies, the detail and implementation of the spatial descriptors vary. Some studies (e.g. [104, 105]) opt to use high-level SIPs such as 'directional loudness maps' or ILD histograms, while others (e.g. [102]) use low-level spatial cues such as ILD and ITD. In short, both approaches are valid and relevant to the RQs presented in this thesis.

Following from this establishment of objective measurements for spatial quality, it is increasingly common practice to use them as additional terms in DL loss functions, thereby allowing the networks to learn spatial features. Erruz [107] demonstrates a loss term based on the ILD of binaural audio, whereas both Lim and Nam [78] and Giampiccolo et al. [108] implement an Inter-channel Level Difference (ICLD) loss term for various combinations of ambisonic channels. In all these studies, the underlying industry motivation was the generation and rendering of new spatial audio formats, similar to the upmixing research mentioned in section 4.1.

4.6 Analysis of large musical datasets

Many of the studies cited in section 4.3 use a large dataset of recorded music to demonstrate the effectiveness of their SIPs. For example, [6, 8, 11] tackle genre recognition, [37] predicts artist/DJ, and [8] determines the decade of the recordings. Similar studies such as [109, 110] show that the geographic

origin of a recording can be determined from audio features, although they don't include any spatial features.

Ziemer et al.'s continuation of their work on goniometer features (see section 4.3) uses a dataset comprising songs that were either produced or rapped by specific artists [111]. They train a Self Organising Map (SOM) on the goniometer features of the dataset. The results firstly show that specific artists have specific sonic profiles, and furthermore that when those artists collaborate, the sound profile of the resulting track is dominated by the producer.

FiXme: These aren't stereo features. Maybe mention that. In Panteli et al.'s work on the classification of singing style [112], the classes – instead of being determined by an a-priori known label – are learned by first extracting low-level audio cues, then finding common features within those cues. The features are then used as the input to unsupervised clustering algorithms to determine the relationships of the recordings to each other. Panteli et al. continued the work in [113] where outliers in the global dataset where found using similar techniques and links between different musical cultures were drawn. On a cautionary note, Chauhan and Pundhir recently showed that ML-based tasks taking audio input from multiple cultures are biased towards Western sources if they use the Mel spectrogram as the audio frontend [114]. The authors recommend using alternative frontends, such as ERB.

Collins et al. collated a dataset of historical electronic music [115]. As part of that work, they extracted two spatial metrics: 'channel loudness difference' and 'stereo spatial ebb'. The example experiment given to demonstrate the dataset showed that 'stereo spatial ebb' changes according to the year of the recording, thus indicating an artistic trend over time. 'Channel loudness difference', on the other hand, remained static.

What all these studies have in common is an interest in the dataset itself regarding both the relationships between individual recordings, and the relationship of the entire corpus to the wider world. Low-level audio features are of lesser importance, other than to construct the high-level metrics that enable comparison between recordings.

5. Research Methodology

In this chapter I break down the reasoning behind the RQs given in section 2.1, both from philosophical and practical standpoints. I argue that the methods I chose for the research in this thesis allow for a timely and achievable answer to the overarching RQ.

The overarching RQ – 'How do music producers use stereo field' – could be approached in many ways. However, the work in this thesis is steeped in the context of large audio datasets. Indeed, the reason for performing the research at the current point in time is the ease of access to said datasets along with the availability of computer processing power to analyse them. From an ontological standpoint this most closely fits the Parmenidean viewpoint of *being*. According to Gray, Parmenidean ontology posits that reality is composed of clearly formed entities with identifiable properties [116]. If the producers' collective musical output is 'reality', then the recordings in the datasets certainly fit that description.

The two smaller RQs represent different epistemological perspectives on the overarching one. RQ_1 is based in *constructivism*, where the meaning of the stereo field is imposed on the recordings by the listener. Conversely, RQ_2 is firmly in the *objectivism* camp, where the properties of a recording's stereo field are an objective truth that can be measured. Not only that, but generalizations about the use of stereo field across the entire population of recordings are also objective and measurable. While the two epistemologies are somewhat adverse to each other, they both fall under the ontology of *being* [117].

Under *constructivism*, one valid research methodology is 'heuristic inquiry' – the act of totally immersing oneself in the phenomenon being researched and allowing intuition to guide the understanding of the RQ. I used this technique for a sizable part of the work described in this thesis. For example, paper I is a retrospective account of building a new musical instrument where the primary sonic material was the signal from a remote stereo microphone. The act of building the instrument, listening to its output, and forming ideas about which sonic phenomena were interesting was a direct way to relate the stereo field with musical knowledge, addressing RQ_1. While this provides a basis for other research, critics of heuristic inquiry will point out that it is deeply personal and therefore not particularly generalizable [116].

To complement the *constructive* approach and form more robust conclusions about an entire population of music producers, I concurrently embarked on a more objective research path. The epistemology of *operationalism* is defined by the principle that in order to understand something, it must be measurable [118]. Thus, to understand stereo field, a set of operations to measure

it is necessary. In practice, that means a software library capable of analysing music recordings and extracting information about their stereo field. The benefits of creating this software are threefold. Firstly, it enables comparative measurements of disparate music recordings, allowing one to validate and challenge the existing research described in chapter 4. Secondly, the methods used for any experimental research must be controllable and repeatable. A software library is a fantastic way to achieve this. Thirdly, the act of writing the library can be seen as a similar act of immersion to that previously described in the *constructive* research. By breaking down the measurement of stereo field into simple instructions, one is forced to think about quantifiable relationships between audio phenomena. Wiltschko argues that language is a tool for both thought and communication in equal measure [119]. It follows that computer code (which is a written language like any other) acts as both a representation of the creator's thought process, and the communication of those thoughts to any readers. The software written for this thesis was therefore fundamental to the research process, as well as underpinning the results of papers III and IV.

One potential criticism of this combinatorial approach is that an answer to the overarching RQ could instead be gained through subjective means, e.g. interviews with music producers. This would quickly run into several problems, however. The first is that of scale. Subjective research is a time-consuming process and the resulting data is deep, but with few examples. The end research results might then not be generalizable to the entire population of music producers. Secondly, the music community holds an inherent understanding that music is a form of language [120]. Adding a layer of indirection (interviews conducted in spoken language) to the research process only serves to muddy the data. Indeed, Hospers argues that an artist's expression is inherently intertwined with the creation the original art form, noting that the intentions are sometimes so complex and multifaceted as to defy explanation even to the artist themselves [121]. In short, it makes most sense to analyse musical output directly.

The work I have performed for this thesis is a combination of immersion in musical practice, thorough reviews of existing literature, and detailed implementation of audio DSP analysis techniques. This combination presents a well-rounded viewpoint on the original RQs, as well as giving rise to new questions that could be answered in the continuation to doctoral studies (see chapter 7).

6. Paper summaries

Paper I

'Development of a Web-Based Real-Time Distributed Reverberation Chamber for Live Concert Applications'

Authors

Austin Franklin, Richard Mitic, Daniel Hedin, Rikard Lindell and Henrik Frisk

Status

Published in *Journal of The Audio Engineering Society*, Mar. 2025 [1].

Research methods

This paper presents a retrospective of a project to build a distributed audio performance over mobile networks. It follows the practice-based research methodology as defined by Candy [122].

Results and conclusions

The paper documents a real use case where stereo field and reproduction thereof is of prime importance. One conclusion was that network reliability was a limiting factor of the audio performance, sometimes leading to a degraded spatial impression. Research question 2.1 is addressed – Finding an adequate balance between audio spatial quality and network utilization would require tools to measure the stereo image.

My contribution

Member of the team that conceptualized the Auxtrument and made the initial survey and prototyping of existing audio transmission technologies. Responsible for post-prototyping of alternative approaches and custom audio software.

Paper II

'A Systematic Literature Review of Methods to Describe the Stereo Image of Recorded Music'

Authors

Richard Mitic and Andreas Rossholm

Status

Accepted to *Journal on Audio, Speech, and Music Processing*, May 2026

Research methods

Following on from the small selection of DSP techniques assessed in paper III, a SLR following the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) standard was performed to gather all currently available methods for determining spatial features of stereophonic musical audio.

Results and conclusions

Addressing research question 2.1, the review revealed several more techniques that can be used to make spatial descriptions of stereo music. Despite the high number of studies that were assessed, there were relatively few 'very relevant' ones. The SLR also revealed that while ML/DL have been used on spatial features in the past, both *unsupervised* ML and very large music corpora remain fairly unexplored. It also details a framework for how to create a SIP algorithm that is useful for any given dataset and use case.

My contribution

Main author, responsible for all data collection and processing, and writing.

Paper IV

'Perceptual Evaluation of Binaurally Panned Speech and Music Mixtures'

Authors

Richard Mitic, Majed Elwardy, Veronica Sundstedt, Andreas Rossholm and Yan Hu

Status

In review at *Acta Acustica*, March 2026.

Research methods

A perceptual test was used to determine listeners' reactions to carefully constructed audio stimuli. Standard statistical analysis was performed on the raw test results, and one particular DSP technique assessed in paper III (PH [13]) was used in an attempt to explain the perceptual results.

Remaining Work

Await peer review and address any comments that come up.

Results and conclusions

The paper concluded that when listeners are presented with voice overlayed on top of music, the best panning position for the voice was not always in the default centre position. The reasons for this were unclear due to a shortage of data, but the paper did demonstrate a real-world use of SIP extraction (using a modified algorithm trialled in paper III). Most relevant to the research question 2.1, it revealed that an objective measurement of panning can show detail that is not easily discernable by listening, such as very slight panning away from the centre.

My contribution

Joint main author alongside Majed Elwardy. Contributed to experiment design, data analysis, software implementation, and writing.

Paper III

'Towards an Objective Comparison of Panning Feature Algorithms for Unsupervised Learning'

Authors

Richard Mitic and Andreas Rossholm

Status

Published in *Proceedings of the 28th International Conference on Digital Audio Effects (DAFx25)*, September 2025

Research methods

Several DSP techniques for determining audio spatial features were performed on a small dataset that had predictable musical characteristics. The different techniques were compared using a new objective metric.

Results and conclusions

The paper revealed that although there are several DSP techniques specifically designed for the extraction of recording and panning features, they are all limited in their own ways. One particular technique (SPF from [6]) was particularly promising as a candidate for use later in the doctoral phase of studies, and the others could be used with some modifications. Overall it highlighted the importance of choosing both a panning feature DSP technique and a distance metric that are appropriate to the overarching task, thereby addressing research question 2.1.

My contribution

Main author, responsible for all data collection, experimental design, and software implementation. Presented the paper at the 28th International Conference on Digital Audio Effects in Ancona, Italy.

7. Discussion and future work

The work documented in this licentiate thesis provides the theoretical and practical basis for analysis of stereo image in music recordings. The focus was on amassing the signal processing techniques that enable the analysis of stereo field, and testing their applicability to various types of recordings that were known a-priori. The next stage of research will involve applying the knowledge from this thesis on much larger musical corpora. The signal processing techniques will be honed and run on vast musical datasets, secure in the knowledge that they will generate data appropriate to the experiment at hand.

In particular, I theorize that examining the SIPs of extremely large musical corpora will reveal patterns in production and recording techniques that were otherwise unobservable. It will also reveal the scale and variation of stereo techniques that musicians are able to employ in their work. Given the basic knowledge gained from the work in this thesis, it might then be possible to reason about how musicians and producers sculpt their stereo fields, relating it both to the musical content and the wider context in which it was created.

During the licentiate, several points arose that provoked further investigation. Firstly, the available public datasets of recorded music are generally encoded using lossy compression. While lossy compression algorithms undergo perceptual tests in their development, it is unclear how they affect spatial cues at the scale of said datasets (which encompass much musical variation). The spatial effects of lossy compression therefore warrant further investigation. Secondly, the public datasets are generally formed from online services where musicians have uploaded their own recordings for free. In contrast, professional recordings (which are much more numerous) are not included in any public datasets. Since professional recordings make up the vast majority of recorded music, it needs to be assessed whether public datasets are representative of professional recordings.

In short, the doctoral research will tackle a larger problem of assessing human musical creativity though the recorded output. The techniques surfaced by the licentiate will provide the foundation, and will be applied to real-world data.

A. Demonstration tracks

These tracks from the FMA dataset [10] were used to generate diagrams of spatial cues and SIPs in chapter 4.

Track ID	Artist	Title	Genres	Letter
123899	Rachel Mason	Rebel Angels	Singer-Songwriter	A
117614	Blue Dot Sessions	Red City Theme	Soundtrack, Instrumental	B
122809	The Carnys	foolsgame	Experimental Pop	C

Glossary

AAC Advanced Audio Coding 36
ADRess Azimuth Discrimination and Resynthesis 37
AGR Automatic Genre Recognition 37
AMT Auditory Modelling Toolbox 31, 32
API Application Programming Interface 14
AR Augmented Reality 27
ASF Amplitude Stereo Features 40, 41
ASW Apparent Source Width 24, 44, 45

BCC Binaural Cue Coding 36
BMS Bivariate Mixture Space 37

CD Compact Disc 29

DAW Digital Audio Workstation 24–26, 29, 35, 42
DJ Disc Jockey 46
DL Deep Learning 11, 43, 44, 46, 51
DSP Digital Signal Processing 13, 16, 18, 36, 43, 49, 51–53

ERB Equivalent Rectangular Bandwidth 32, 33, 40, 47

FA Frequency-Azimuth 37, 39
FFT Fast Fourier Transform 30, 32, 40
FMA Free Music Archive 11, 35, 36, 39–41, 55

HRIR Head-Related Impulse Response 21, 22

IACC Interaural Cross Correlation 18, 23, 37, 44, 45
ICLD Inter-channel Level Difference 46
IIR Infinite Impulse Response 33
ILD Interaural Level Difference 18–21, 23, 37, 40, 44, 46
IPD Interaural Phase Difference 18–20
ITD Interaural Time Difference 18, 19, 21, 23, 44, 46

LEF Lateral Energy Fraction 45
LEV Listener Envelopment 24, 44, 45
LFE Low Frequency Effects 27, 28

MIR Music Information Retrieval 11, 13, 37

ML Machine Learning 11, 32, 36, 47, 51
MP3 MPEG layer 3 36
MPEG Moving Picture Experts Group 36

PC Panning Coefficients 40
PH Panning Histogram 39–41, 52
PI Panning Index 37
PRISMA Preferred Reporting Items for Systematic reviews and Meta-Analyses 51
PS Parametric Stereo 36
PSF Phase Stereo Features 41

RMS Root Mean Squared 19
RQ research question 13, 14, 35, 46, 48, 49

SIP Stereo Image Profile 14, 38–40, 45, 46, 51, 52, 54, 55
SLR Systematic Literature Review 14, 15, 51
SOM Self Organising Map 47
SPF Stereo Panning Features 38–41, 53
SPL Sound Pressure Level 19
SSPS Spectral Stereo Phase Spread 41
STFT Short-time Fourier Transform 30, 31, 34, 40, 41

TF time-frequency 30, 31, 34, 37

VR Virtual Reality 27

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